

An Industry-Oriented Mini Project report
on

TOOLS FOR EASY MANAGEMENT OF FILES AND DIRECTORIES

Submitted in partial fulfillment of the Academic
requirement of the Award of Degree of

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In

COMPUTER SCIENCE AND ENGINEERING (AIML)

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Kandlakoya , Medchal Road, Hyderabad .

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to JNTUH, Hyderabad, in partial fulfillment of the requirement for award of the degree of B.Tech in CSE and is a record of a bonafide work carried out under our guidance and supervision. The results in this project have been verified and are found to be satisfactory. The results embodied in this work have not been submitted to have any other University for award of any other degree or diploma

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ABSTRACT

Many of the organizations connect plenty of numbers of systems to establish a network which will intern make their work easier to share their folders and files. As many systems are involved security concern is the major aspect while attaching such systems, and wanted to keep track of the network system activities for security motive. A Monitoring mechanism in a computer grid is used to observe all the ongoing activities of the whole network. The main objective is to gather details from the monitoring environment and the system. In this paper, various network parameters of the established computer networks are observed using the developed monitoring mechanism such as IP address, files transferred and Mac address.

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INTRODUCTION

GENERAL INTRODUCTION

A large number of the organization connects plenty of numbers of systems to build a network because of which they can easily share their files and folders. Multiple systems security are being connected and privacy plays a major role, so wanted to monitor all the ongoing network activities wanted to monitored for security purpose. Networks can be monitored by using Wireshark [8] and scrutinizing mechanism to precisely determine the utilization of various resources, traffic flows and performance-related parameters on a network. The prime characteristic of effective monitoring tool is that it must provide both graphical and numerical representations of the system at any point of time[1]. Monitoring tools are important programs to verify the wellbeing of the network. The main task of the network monitoring is to continuously monitor computer network for breakdown or failing part of the system and it triggers a notification to the network administrator under any of such circumstances. Network monitoring includes numerous methods which are installed to preserve the integrity and security of the network. To monitor the system behavior following parameters are identified Host ID, IP address, Network ID, File Name, Memory Utilization, Process Utilization and Bandwidth Utilization, Network monitoring systems and intrusion detection system are two mutually exclusive systems. The advantages of having a network monitoring tool [2].

Reliability: Network monitoring tool continuously monitors and keep track of a crucial part of system or software if there is any malicious activity it notifies network administrator before it escalates.

Stay in the know: If there is any failure event, a monitoring system will be able to send an alert message or signal to the computers or mobile device of the network administrator.

Troubleshooting: When an effective network monitoring tool is in place, the part of the system can be quickly recognized that is originating the issue can be quickly recognized, thereby can be cut short the downtime and no much time is needed to diagnose the problem.

SYSTEM PROPOSAL

EXISTING SYSTEM :

The entire work is divided into 5 stages. Collecting information regarding the distributed systems and then building it is done in the first stage. The architecture of a distributed system consists of workstations collection and server connected by a Local Area Network. Monitoring the distributed file sharing is done in the second stage. This monitoring is responsible for collecting and analyzing the different issues that arise in distributed systems by having a centralized monitor. The several parameters of monitoring address the questions such as what can be monitored, how can be monitored and what level of details to be monitored [9]. In the third stage, the information collected through monitoring is stored in a centralized database. So that the information can be re-used for later usage if needed. In the fourth stage, external hacker such as Wireshark[8] has been used to check how secure the proposed distributed system is. In the final stage, security metrics are calculated to determine the secure transferring of files between the user's system and monitoring system.

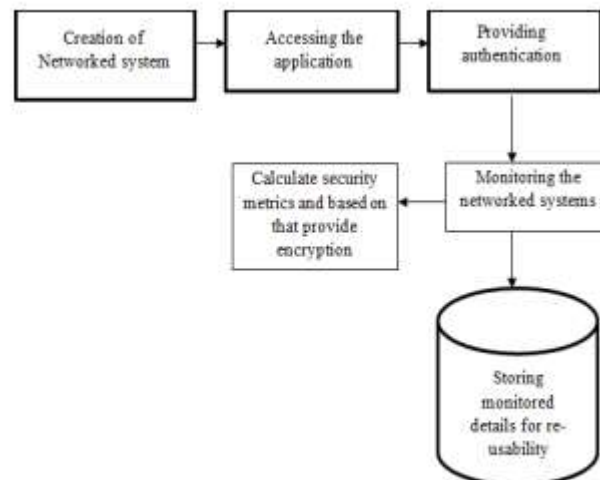


Fig. 1. Proposed System

The system architecture mainly contains 2 modules namely Admin module and Client module. In the admin module, an admin can authorize registered users and also he can monitor the activities of every user. Admin has the option to accept or reject the registered nodes. A client can access the network only if admin approves. Admin can view activities of each user such as data communication, name of the file transferred and message. He can set the criteria as well as he can store monitored details and log details such as IP address, events taken place with date and time [10]. The client module is for end-users. The end-user has to register first and then can log in to the system with the credentials. Once after logging in, the user can create new messages, can view sent messages and can

view received messages. CPU utilization, memory utilization, disk utilization and bandwidth utilization is stored in the database when closing of the application.

Security Metric Calculation: The security issues and metrics are calculated in terms of the level of criticality, detail and size.

A.Criticality of data: This attribute gives the criticality details of data with respect to security [12]. In other words, this attribute is directly related to deciding the level of security. Higher the rate of criticality, the larger the value of security metric.

B.Detail of data:

This attribute gives the details of the level of concreteness/abstraction of monitored data. E.g. monitoring the communication, one can be interested only in monitoring the less detailed information such as the number of texts exchanged between the objects or one can be interested in monitoring detailed information such as the texts and its contents. Higher the rate of detailed data, the larger the value of security metric for the attribute.

C.Size of data This attribute deals with the measuring of the amount of data collection during monitoring[11]. This is one of the important attributes.

E.g. the monitoring may cause network congestion in the target system, which in turn holds back the quality of services to be delivered. To ascertain the influence of monitoring on the efficiency of security appliances of the target machine, some metrics need to be elucidated. Based on the set of attribute functions only metrics needs to be decided accordingly data be collected by a monitoring system. This will intern influence security implementation. The following attributes are pinpointed as relevant to the security concerns, and the metrics are classified in terms of these attributes: level of Criticality, Detail, and Size[3].

- ☐ The criticality of data is a feature that represents the significance of the data with respect to security. The data criticality feature is directly proportional to the security metrics.
- ☐ The level of distraction of monitored data is decided by the details of the collected attributes.
- ☐ The size feature represents the amount of data composed during monitoring. This attribute is of prime concern as it represents the likelihood of network congestion in the target machine.

The security metrics, SM, can be defined by the equation:

$$SM = \alpha.C + \beta.D + \gamma.S \text{ ————— (1)}$$

Where

C is the level of security criticality attribute of the data

D is the level of the detail of the data.

S is the size of the data.

α , β , γ are non-negative coefficients whose values and relationships can be determined using some analytical techniques.

Before calculating the value of α , β , γ the following values need to be assumed. Three attributes of each parameter can be considered out of 1 weight and the assumed values are listed in the following tables.

Value	Level of Criticality Meaning
0	Not critical
0.25	Less critical
0.50	Medium
0.75	More critical
1.0	Extremely critical

Table 2 Assumed values for the level of detail.

Value	Level of detail Meaning
0	Not detailed
0.25	Less critical
0.50	Medium detailed
1.0	Fully detailed

Table 3 Assumed values for the size of data as below

Value	Size of data Meaning
0.1	Size is less
1.0	Size is more

After measuring the security metric values, the parameters having the highest value is considered to be the more critical parameter and encryption should be applied to it. A network packet analyzer tool like Wireshark [8] attempts to intercept a network packet and tries as deeply as possible to view the packet data. To capture the packets, an interface have to be selected from the capture menu. At the top of our network, packets are shown with information such as source , destination, protocol length, data. Right click on any TCP packet to select "Follow TCP pipe" to see the data in the packet stream.

DISADVANTAGES

- Here we don't use any algorithms.
- Here the working is having less accuracy rate.
- Less security.

PROPOSED SYSTEM :

The following three parameters are for measuring the value of α , β , γ are taken into account and the following assumptions are made. The boundaries considered for ascertaining the estimation of α , β , γ are IP address, Network ID, and Host ID. At first, the security measurements for every boundary is considered as 1 and α , β , γ for IP address is 2, 1, 1, α , β , γ for Network ID is 1, 1.5, 0.5 and α , β , γ for Host ID is 0.5, 1.5, 1 individually. From the systematic technique the accompanying qualities are acquired for α , β , γ i.e., $\alpha=0.6668$, $\beta=1.222$, $\gamma=0.666$. In the wake of applying α , β , γ values, the following qualities are acquired for security measurements.

Table 4 Security metric values for different parameters

Parameters	Security Metrics
File Content	2.667
Filename	0.90004
Host ID	0.40003
IP Address	2.06688
Network ID	1.73348
CPU Utilization	1.56678
Bandwidth Utilization	1.56678
Memory	1.56678

After the Security Metrics value has been measured and analyzed, the 'File Content' Factor is found to have a high value, so more essential parameter is considered and security techniques are applied.

The 'File Material' only consists of Text details. Client-side data, just before submitting messages or files File or file encryption is achieved using the AES algorithm.

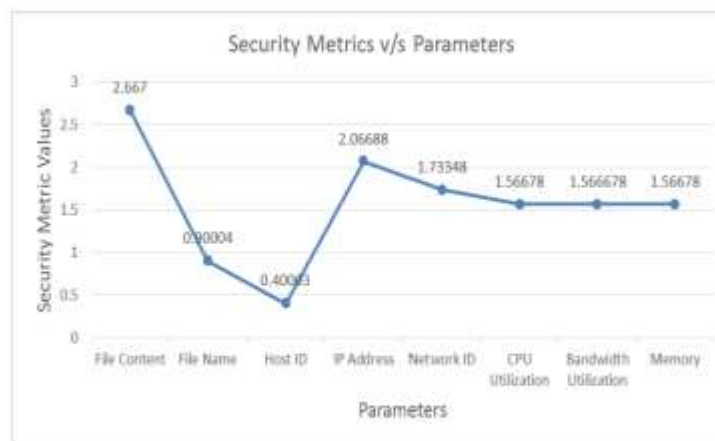


Fig.3. Representation of Security metrics value.

Download time for a single system: The graph is plotted for the download time and the files of size varying from 1MB to 1GB. From the graph, can be concluded that as the file size increases the download time also increases.

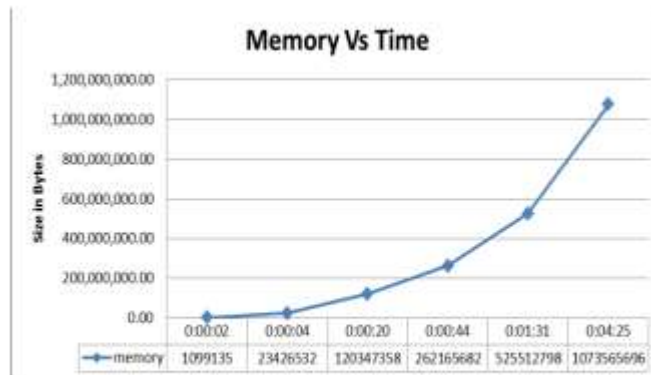


Fig.4. Download time for a single system

Download Time For Varying Number Of Systems: The graph is plotted for the download time and a varying number of systems. A file of 100MB is downloaded by considering one to four systems. From the graph, can be concluded that as the number of systems increases the download time also increases.

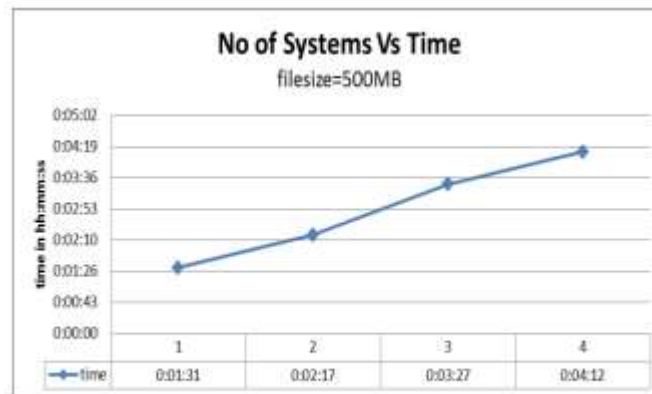


Fig.5. Download time for a varying number of systems

ADVANTAGES

- Here we are using AES algorithm.
- It is more efficient and secured.

LITERATURE SURVEY :

Natascha Petry Ligocki et.al[1] proposed System checking is a difficult process because for the most part requires the utilization of various instruments for explicit purposes. To intended to reach a wide scope of checking needs system observing device, called PaQueT is mentioned in this paper. The client can characterize measurements as questions in a procedure like composing inquiries on a database the executive's framework. PaQueT permits one to screen esteems going from bundle level measurements to those generally gave uniquely by devices dependent on SNMP. PaQueT has been created as an augmentation of the Borealis Data Stream Management System. The main bit of scope of this methodology is the capacity to produce estimations continuously, limiting the volume of information put away; second, the instrument can be handily stretched out to think about a few kinds of system conventions. They have directed a test study to check the viability of our methodology and to decide its ability to process enormous volumes of information. organize checking apparatus called Pa-QueT (Packet Query Tool), which has been intended to cover a wide scope of observing needs.

Brajesh Pande et.al[2] developed the system called: —Pick Packet filter. The broad utilization of systems for a trade of data has additionally had repercussions on the development and spread of crime through their utilization. Law requirement organizations need to stay aware of the developing patterns in these zones for identification and anticipation. Among the few needs of such offices is the need to screen, identify and investigate bothersome system traffic. Nonetheless, observing, identifying, and examination of this traffic might be against the objective of keeping up the protection of people whose arrange correspondences are being checked. Pick Packet is a system checking apparatus that handles the clashing issues of system observing and protection through its reasonable use. PickPacket has four segments "The PickPacket Configuration File Generator" for helping the client in setting up the boundaries for catching bundles, the "PickPacket Packet Filter" for catching parcels, the "PickPacket Post-Processor" for investigating parcels, and the "PickPacket Data Viewer" for demonstrating the caught information to the client. Sniffers that can dump information in the wake of looking at the substance of an applicationlevel parcel are required for ensuring the security of people just as to help legitimate observing of the system. PickPacket is a valuable apparatus for checking data streaming over the system. The plan of PickPacket Filter is measured, adaptable, extensible, and productive. In-piece sifting makes the instrument quick and the utilization of the Boyer Moore calculation for text string look through makes the apparatus quicker. Reasonable utilization of the framework can help secure the protection of people and can dump just important information to the circle.

Wonchul Kang et.al[3] proposed a simple method of monitoring the network. Ongoing conversations on the progressive Future Internet engineering have prompted Content-

Centric Networking (CCN) that puts an accentuation on substance for the effective data conveyance. In this paper, they propose a simple method of monitoring the network for CCN based on SNMP. From IPFIX, they could easily find detailed content information provided in interest or data packets in flow format, while retrieving information in SNMP about the CCN node and its tables.

The CCN is an opensource software project and it's implementation supports IP to test and improve new CCN features or applications. There are two significant packet types in CCN: Interest and Information. An Interest packet will contain the service name requested by clients or CCN nodes, and the corresponding service will be received in Data packets sent by origin servers or intermediate CCN nodes.

Silva et al. [4] developed an object-oriented model that is composed of three packages, monitoring package is responsible for monitoring the different entities such as process utilization of various hosts in the distributed system and it informs to event detection and notification package whenever there is a change in the value of collected parameters. The event detection and notification package are responsible for analyzing data and determining the occurrence of events.

Hiltunen and Schlichting[5] designed a model for an adaptive distributed system which includes three phases namely Change detection, Agreement and Action. Change detection is responsible for monitoring the possible changes in the environment and decides when that change has occurred. The agreement is responsible for reaching an agreement among

all sites that adaptation is required and the Action phase is responsible for changing the behavior of the system based on agreed changes.

Rajinder Singh et.al[6] suggested an easy technique of monitoring the network by using packet sniffers. Mainly they concentrated on factors such as exploration and analyzation of the network issues, misemploying the network by intrinsic and extrinsic users and observing network bandwidth exertion.

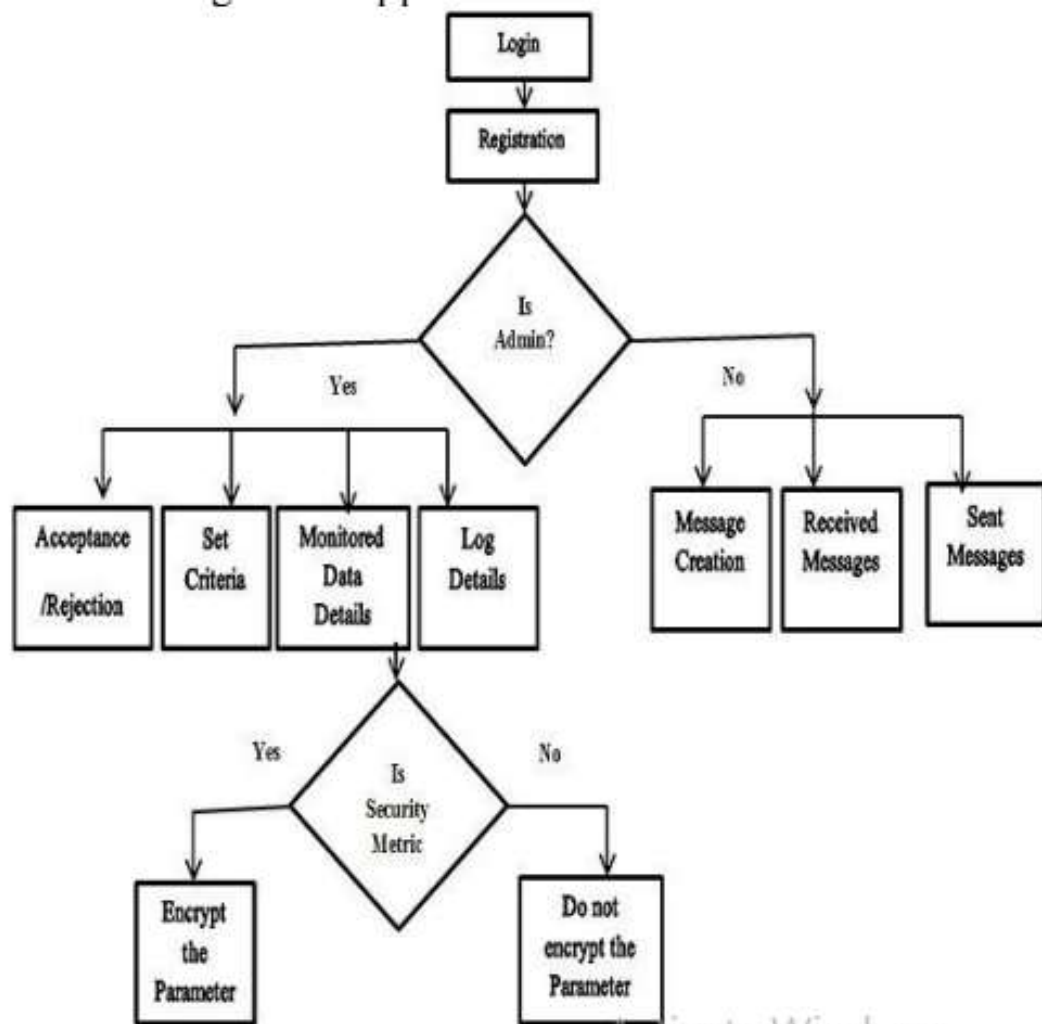
Client connectivity issues are addressed with help of signal strength collected by the client machine, Capacity of data transference was modeled with help of channel selection algorithm which used to select a channel with less interference. The packet snipping mechanisms considered here are unlocked and easily accessible on the World Wide Web. In this paper wireless debugging issues such as internet connectivity and data crowding issues are discussed.

Yu Liu et.al [7] explained about the localization of algorithms for the wireless network. The proposed approach is detailed as a relapse issue, which is understood by Elastic Net. Not at all like different past limitation draws near, the proposed approach beats the deficiencies of conventional methodologies accept that hubs are conveyed in standard zones without gaps

or deterrents, in this manner has solid flexibility to the mind-boggling sending condition. An epic multi-jump calculation for a remote organization with unevenly appropriated hubs is proposed in this paper. The proposed approach depends on the flexible net, which is a regularization technique with great interpretability and security. The proposed approach applies to the scenarios where nodes are unevenly distributed.

SYSTEM DIAGRAM

ARCHITECTURE DIAGRAM



Activate Windows
Go to Settings to activate Windc

UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

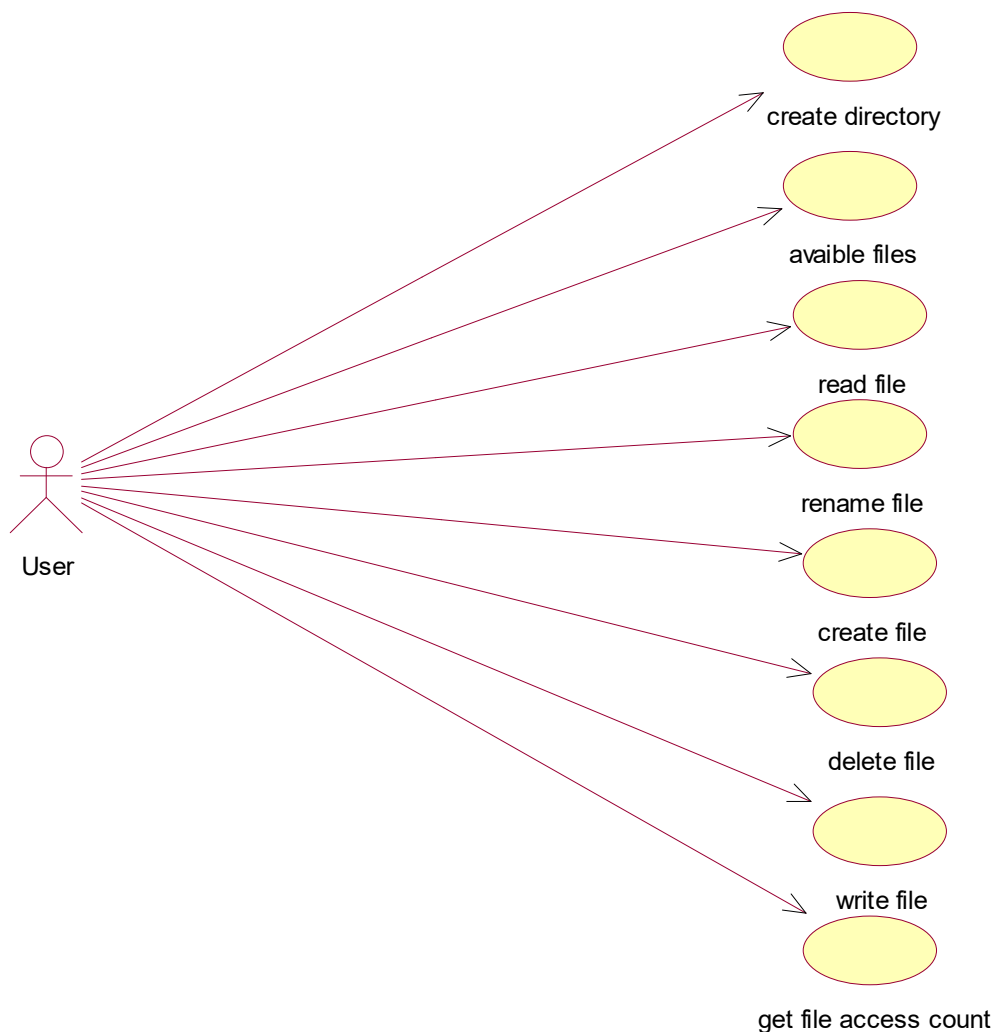
GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.
7. Integrate best practices.

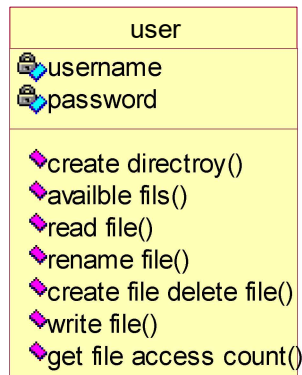
USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



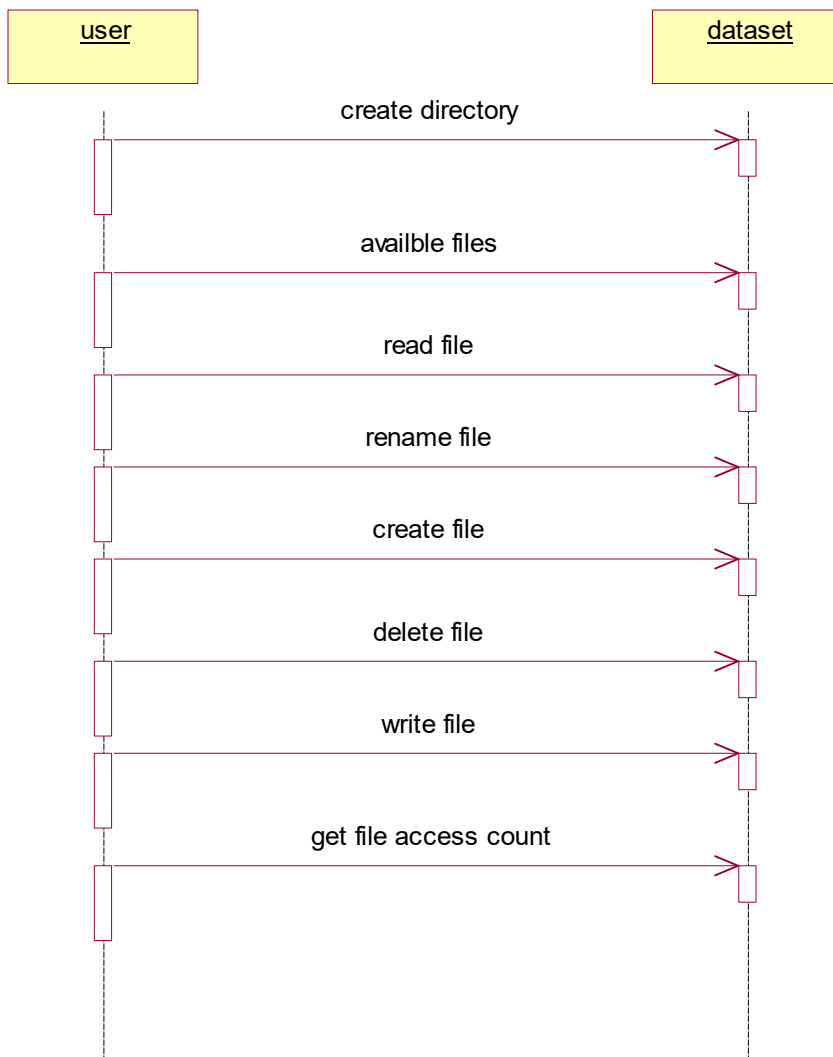
CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



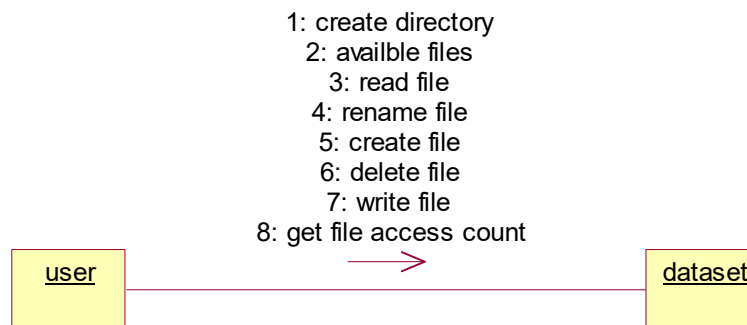
SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



COLLABRATION DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.



IMPLEMENTATION

MODULES:

To implement this project we have designed following modules

1. New User Signup Here: button to signup with the application and get below output
2. Create Directory: button to enter directory name and get below output.
3. project_files: and press OK to create directory and get below output.
4. Create File: button to create file.
5. Read File : see some text is written to file and we can read that data back by clicking on 'Read File' button .
6. Get File Access Count: button to get access count of selected file.

MODULE DESCRIPTION

1. New User Signup Here:

- Allows new users to register with the application.
- Collects user information and stores it securely for future access.
- Provides a way for users to create a unique account.
- Validates and verifies user input for accurate sign-up.
- Generates unique identifiers or credentials for user identification.
- Enables a seamless onboarding process for new users.
- Integrates error handling to ensure smooth registration.
- Offers essential security measures for user data protection.
- Establishes a foundation for personalized user experiences.
- Facilitates interaction with the application through user accounts.

2. Create Directory:

- Enables users to define a new directory within the application.
- Collects the desired directory name from the user.
- Validates the directory name to meet system requirements.
- Initiates the creation of the specified directory.
- Provides feedback on successful directory creation.
- Integrates error handling for any issues during directory creation.
- Offers a straightforward way to organize and manage files.
- Enhances the structure and order of the file system.
- Encourages efficient file organization for users.
- Facilitates better management of digital content.

3. project_files:

- Represents a module or function for handling project-related files.
- Interacts with the user to initiate specific project actions.

- Supports the creation, management, and storage of project-related files.
- Works in tandem with other modules to streamline project processes.
- Enhances project organization and accessibility for users.
- Provides a central point for project file management.
- Integrates with the directory creation module for structured project storage.
- Ensures efficient access and retrieval of project files.
- Supports collaboration and teamwork on project-related tasks.
- Contributes to a cohesive project management experience.

4. Create File:

- Allows users to create new files within the application.
- Collects file-related information such as name and content from the user.
- Validates user input and ensures proper file creation parameters.
- Initiates the creation of the specified file with provided content.
- Provides feedback on successful file creation.
- Integrates error handling for any issues during file creation.
- Enables users to generate and organize content effectively.
- Supports various file types and content structures based on user needs.
- Facilitates a structured approach to file creation and management.
- Enhances the overall file handling capabilities of the application.

5. Read File:

- Allows users to view the content of a selected file.
- Reads and displays the text or data stored within the chosen file.
- Provides a simple interface for users to access file contents.
- Enables quick and convenient review of file information.
- Supports better understanding of file data and purpose.
- Integrates error handling for cases where file reading is not possible.
- Promotes efficient usage and extraction of data from files.
- Facilitates better decision-making based on file content.
- Encourages users to interact with stored data in the application.

6. Get File Access Count:

- Allows users to retrieve the access count for a specific file.
- Tracks and maintains a count of times a file has been accessed.
- Provides insights into file popularity and usage patterns.
- Enhances user awareness of file access frequency.
- Supports analytics and decision-making based on access counts.
- Integrates error handling for cases where access count retrieval fails.
- Enables a deeper understanding of user engagement with files.
- Facilitates monitoring and analysis of file usage trends.
- Encourages users to optimize their file usage based on access data.

SYSTEM REQUIREMENTS

HARD REQUIRMENTS :

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Floppy Drive : 1.44 Mb.
- Monitor : 15 VGA Colour.
- Mouse : Logitech.
- Ram : 512 MB.

SOFTWARE REQUIRMENTS :

- Operating system : Windows 7 Professional.
- Coding Language : python

TESTING OF PRODUCTS

SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successful unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.
- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.
- Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

SOURCE CODE

```
from tkinter import messagebox
from tkinter import *
from tkinter import simpledialog
import tkinter
from tkinter import ttk
import pymysql
import pickle
import shutil
import os

main = tkinter.Tk()
main.title("Tool for Easy Management of Files and Directories")
main.maxsize(width=500 , height=300)
main.minsize(width=500 , height=300)

global login_user, login_pass, sign_user, sign_pass, contact, username, count,
winsignup, winlogin, text, dirname, filecombo, accessList
count = 0
global available_files
access_count = {}

def createDirectory():
    global username, text
    global dirname
    dirname = simpledialog.askstring(title="Please enter directory name",
prompt="Please enter directory name")
    if os.path.exists('files/'+username) == False:
        os.mkdir('files/'+username)
    if os.path.exists('files/'+username+"/"+dirname) == False:
        os.mkdir('files/'+username+"/"+dirname)
    text.insert(END,"Directory Created\n")

def createFile():
    global username, text
    global dirname
    dirname = simpledialog.askstring(title="Please enter directory name",
prompt="Please enter directory name")
    filename = simpledialog.askstring(title="Please enter file name", prompt="Please
enter file name")
    if os.path.exists('files/'+username+"/"+dirname):
        open('files/'+username+"/"+dirname+"/"+filename, 'w').close()
        text.insert(END,"File Created: "+filename)
        available_files.append('files/'+username+"/"+dirname+"/"+filename)
        filecombo['values'] = available_files
```

```

else:
    text.insert(END, "Given path does not exists\n")

def deleteFile():
    global username, text
    global dirname
    dirname = filecombo.get()
    dirname = dirname.replace("\\", "/")
    arr = dirname.split("/")
    usernames = arr[1]
    dirname = arr[2] + "/" + arr[3]
    if os.path.exists('files/' + username + "/" + dirname):
        if os.path.isdir('files/' + username + "/" + dirname):
            shutil.rmtree('files/' + username + "/" + dirname)
        if os.path.isfile('files/' + username + "/" + dirname):
            os.remove('files/' + username + "/" + dirname)
        text.insert(END, "Given file deleted\n")
        available_files.remove(filecombo.get())
        filecombo['values'] = available_files
    else:
        text.insert(END, "file does not exists\n")

def writeFile():
    global username, text
    global dirname
    dirname = filecombo.get()
    dirname = dirname.replace("\\", "/")
    arr = dirname.split("/")
    current_name = arr[1]
    dirname = arr[2]
    filename = arr[3]
    filetext = simpledialog.askstring(title="Please enter file content", prompt="Please
enter file content")
    if os.path.exists('files/' + username + "/" + dirname + "/" + filename):
        f = open('files/' + username + "/" + dirname + "/" + filename, "a")
        f.write(filetext + "\n")
        f.close()
        text.insert(END, "file data saved at server\n")
    else:
        text.insert(END, "file does not exists\n")

def readFile():
    global username, text
    global dirname
    text.delete('1.0', END)
    dirname = filecombo.get()
    dirname = dirname.replace("\\", "/")

```



```

arr = dirname.split("/")
usernames = arr[1]
dirname = arr[2]
filename = arr[3]
if os.path.exists('files/'+username+"/"+dirname+"/"+filename):
    with open('files/'+username+"/"+dirname+"/"+filename) as f:
        data = f.read()
    f.close()
    text.insert(END,"File Content Showing in Below lines\n\n")
    text.insert(END,data)
    if filename in access_count.keys():
        access_count[filename] = access_count.get(filename) + 1
    else:
        access_count[filename] = 1
else:
    text.insert(END,"files does not exists\n")

def renameFile():
    global username, text
    global dirname
    dirname = filecombo.get()
    dirname = dirname.replace("\\", "/")
    arr = dirname.split("/")
    usernames = arr[1]
    dirname = arr[2]
    oldname = arr[3]
    newname = simpledialog.askstring(title="Please enter new file name", prompt="Please
enter new file name")
    if os.path.exists('files/'+username+"/"+dirname+"/"+oldname):
        os.rename('files/'+username+"/"+dirname+"/"+oldname, 'files/'+username+"/"+dirna
me+"/"+newname)
        text.insert(END,"file rename successfully\n")
    else:
        text.insert(END,"file does not exists\n")
    available_files.append('files/'+username+"/"+dirname+"/"+newname)
    for i in range(len(available_files)):
        ff = available_files[i]
        if os.path.basename(ff) == oldname:
            available_files.remove(ff)
            break
    filecombo['values'] = available_files
    text.insert(END,"File renamed successfully\n")

def accessCount():
    text.delete('1.0', END)
    for name,count in access_count.items():

```

```

        text.insert(END, "File Name = "+name+" Access Count = "+str(count)+"\n")

def fileSystem():
    global username, text, available_files, filecombo, accessList
    fs = tkinter.Tk()
    fs.title("Tool for Easy Management of Files and Directories")
    fs.maxsize(width=1300 , height=900)
    fs.minsize(width=1300 , height=900)
    font1 = ('times', 13, 'bold')

    cdButton = Button(fs, text="Create Directory", command=createDirectory)
    cdButton.place(x=50,y=100)
    cdButton.config(font=font1)

    cfButton = Button(fs, text="Create File", command=createFile)
    cfButton.place(x=300,y=100)
    cfButton.config(font=font1)

    filecombo = ttk.Combobox(fs, values=available_files, postcommand=lambda:
filecombo.configure(values=available_files))
    filecombo.place(x=50,y=150)
    if len(available_files) > 0:
        filecombo.current(0)
    filecombo.config(font=font1)

    dfButton = Button(fs, text="Delete File", command=deleteFile)
    dfButton.place(x=300,y=150)
    dfButton.config(font=font1)

    rfButton = Button(fs, text="Read File", command=readFile)
    rfButton.place(x=50,y=200)
    rfButton.config(font=font1)

    wfButton = Button(fs, text="Write File", command=writeFile)
    wfButton.place(x=300,y=200)
    wfButton.config(font=font1)

    renButton = Button(fs, text="Rename File", command=renameFile)
    renButton.place(x=50,y=250)
    renButton.config(font=font1)

    renButton = Button(fs, text="Get File Access Count", command=accessCount)
    renButton.place(x=300,y=250)
    renButton.config(font=font1)

    text=Text(fs,height=15,width=120)

```

```

scroll=Scrollbar(text)
text.configure(yscrollcommand=scroll.set)
text.place(x=10,y=300)
text.config(font=font1)

fs.config(bg='chocolate1')
fs.mainloop()

def readFiles():
    global username, available_files
    if len(available_files) > 0:
        available_files.clear()
    available_files.append("Available Files")
    for root, dirs, directory in os.walk('files/'+username):
        for j in range(len(directory)):
            available_files.append(root+"/"+directory[j])

def validateLogin():
    global login_user, login_pass, username, available_files
    available_files = []
    usr = login_user.get()
    password = login_pass.get()

    output = "none"
    con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = '',
database = 'distributed',charset='utf8')
    with con:
        cur = con.cursor()
        cur.execute("select username, password FROM register")
        rows = cur.fetchall()
        for row in rows:
            if row[0] == usr and row[1] == password:
                output = "success"
                username = usr
                readFiles()
                break
    if output == "success":
        winlogin.destroy()
        fileSystem()
    else:
        messagebox.showinfo("Login Failed. Please Retry")

def signupAction():
    global sign_user, sign_pass, contact, username, count, winsignup
    usr = sign_user.get()

```

```

password = sign_pass.get()
contactno = contact.get()

output = "none"
con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = '',
database = 'distributed',charset='utf8')
with con:
    cur = con.cursor()
    cur.execute("select username FROM register")
    rows = cur.fetchall()
    for row in rows:
        if row[0] == usr:
            output = username+" Username already exists"
            break
    if output == "none":
        db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root',
password = '', database = 'distributed',charset='utf8')
        db_cursor = db_connection.cursor()
        student_sql_query = "INSERT INTO register(username,password,contact)
VALUES('"+usr+"','"+password+"','"+contactno+"')"
        db_cursor.execute(student_sql_query)
        db_connection.commit()
        print(db_cursor.rowcount, "Record Inserted")
        if db_cursor.rowcount == 1:
            output = "Signup process completed. You can login now"
            count = 2
            messagebox.showinfo(output,output)
            winsignup.destroy()
            loginFunction()
    else:
        messagebox.showinfo(output,output)

def loginFunction():
    global login_user, login_pass, count, winlogin
    if count == 0:
        main.destroy()
        count = 1
    winlogin = tkinter.Tk()
    winlogin.title("User Login Screen")
    winlogin.maxsize(width=500 , height=300)
    winlogin.minsize(width=500 , height=300)

    l1 = Label(winlogin, text='Login Screen')
    l1.config(font=font1)
    l1.place(x=140,y=30)

```

```

l2 = Label(winlogin, text='Username')
l2.config(font=font1)
l2.place(x=50,y=80)

login_user = Entry(winlogin,width=35)
login_user.config(font=font1)
login_user.place(x=150,y=80)

l3 = Label(winlogin, text='Password')
l3.config(font=font1)
l3.place(x=50,y=130)

login_pass = Entry(winlogin,width=35, show="*")
login_pass.config(font=font1)
login_pass.place(x=150,y=130)

login1Button = Button(winlogin, text="Submit", command=validateLogin)
login1Button.place(x=100,y=180)
login1Button.config(font=font1)

winlogin.mainloop()

def signupFunction():
    main.destroy()
    global sign_user, sign_pass, contact, winsignup
    winsignup = tkinter.Tk()
    winsignup.title("New User Signup Screen")
    winsignup.maxsize(width=500 , height=400)
    winsignup.minsize(width=500 , height=400)

    l1 = Label(winsignup, text='New User Signup Screen')
    l1.config(font=font1)
    l1.place(x=140,y=30)

    l2 = Label(winsignup, text='Username')
    l2.config(font=font1)
    l2.place(x=50,y=80)

    sign_user = Entry(winsignup,width=35)
    sign_user.config(font=font1)
    sign_user.place(x=150,y=80)

    l3 = Label(winsignup, text='Password')
    l3.config(font=font1)
    l3.place(x=50,y=130)

    sign_pass = Entry(winsignup,width=35, show="*")

```

```

sign_pass.config(font=font1)
sign_pass.place(x=150,y=130)

l4 = Label(winsignup, text='Contact No')
l4.config(font=font1)
l4.place(x=50,y=180)

contact = Entry(winsignup,width=25)
contact.config(font=font1)
contact.place(x=150,y=180)

sign1Button = Button(winsignup, text="Submit", command=signupAction)
sign1Button.place(x=100,y=230)
sign1Button.config(font=font1)

winsignup.mainloop()

def closeFunction():
    main.destroy()

font = ('times', 16, 'bold')
title = Label(main, text='', anchor='w')
title.config(bg='darkviolet', fg='gold')
title.config(font=font)
title.config(height=3, width=120)
title.place(x=0,y=5)

font1 = ('times', 12, 'bold')

loginButton = Button(main, text="Login Here", command=loginFunction)
loginButton.place(x=100,y=100)
loginButton.config(font=font1)

loginButton = Button(main, text="New User Signup Here", command=signupFunction)
loginButton.place(x=100,y=150)
loginButton.config(font=font1)

exitButton = Button(main, text="Exit", command=closeFunction)
exitButton.place(x=100,y=200)
exitButton.config(font=font1)

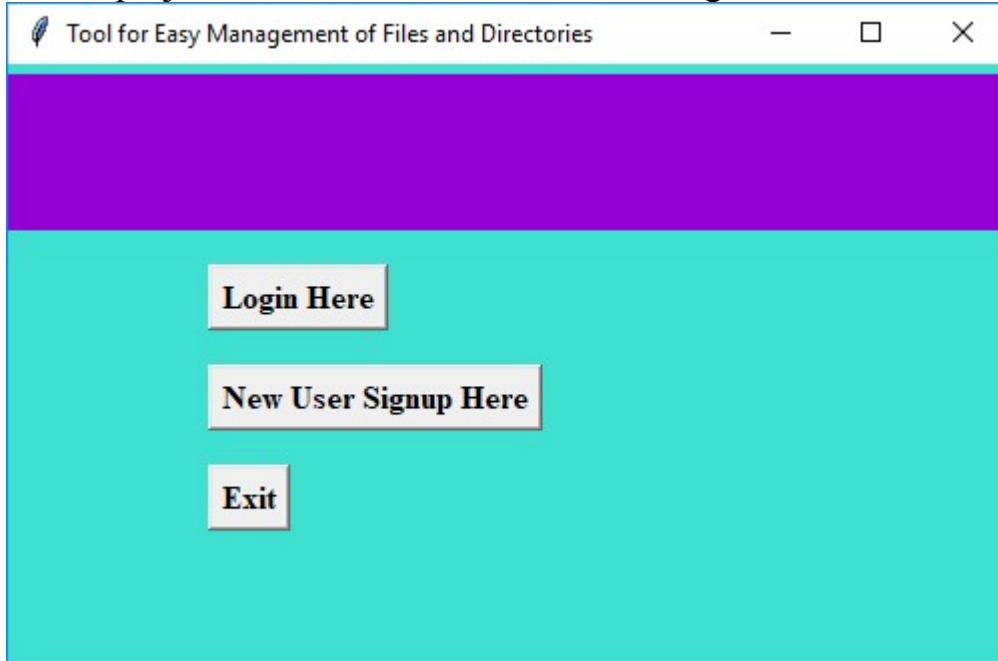
main.config(bg='turquoise')
main.mainloop()

```

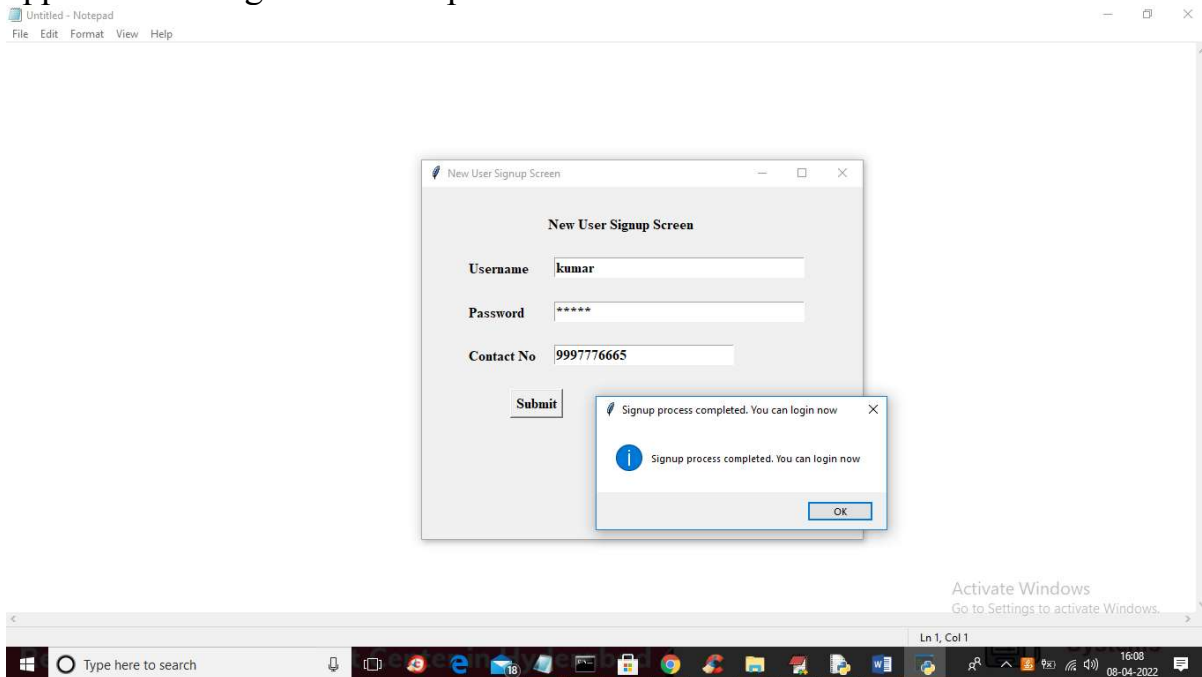
SAMPLE SCREENSHOTS

SCREENSHOTS :

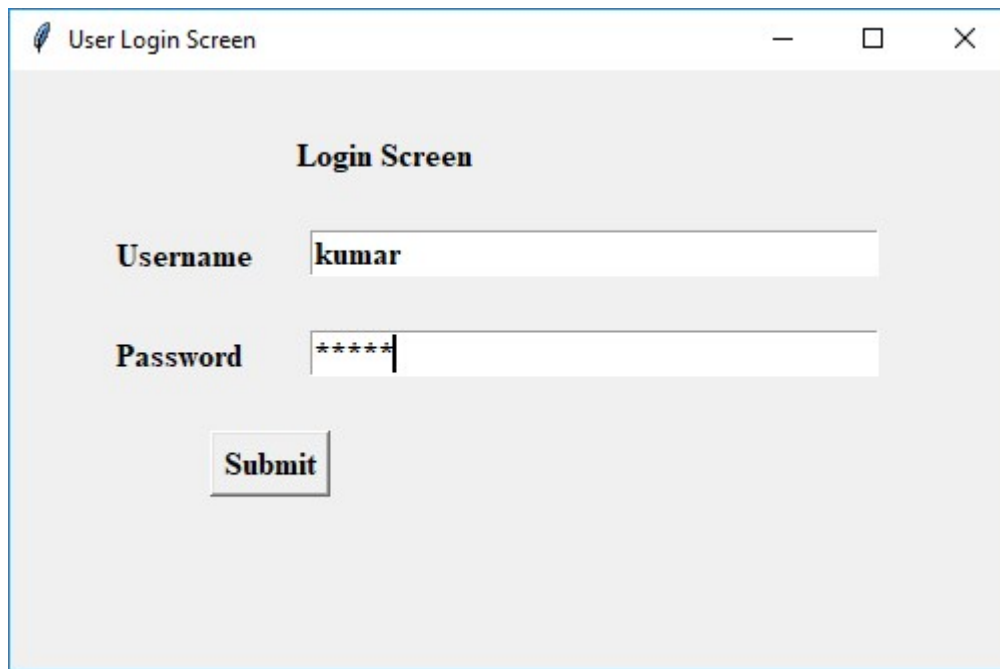
To run project double click on 'run.bat' file to get below screen



In above screen user has to click on 'New User Signup Here' button to signup with the application and get below output

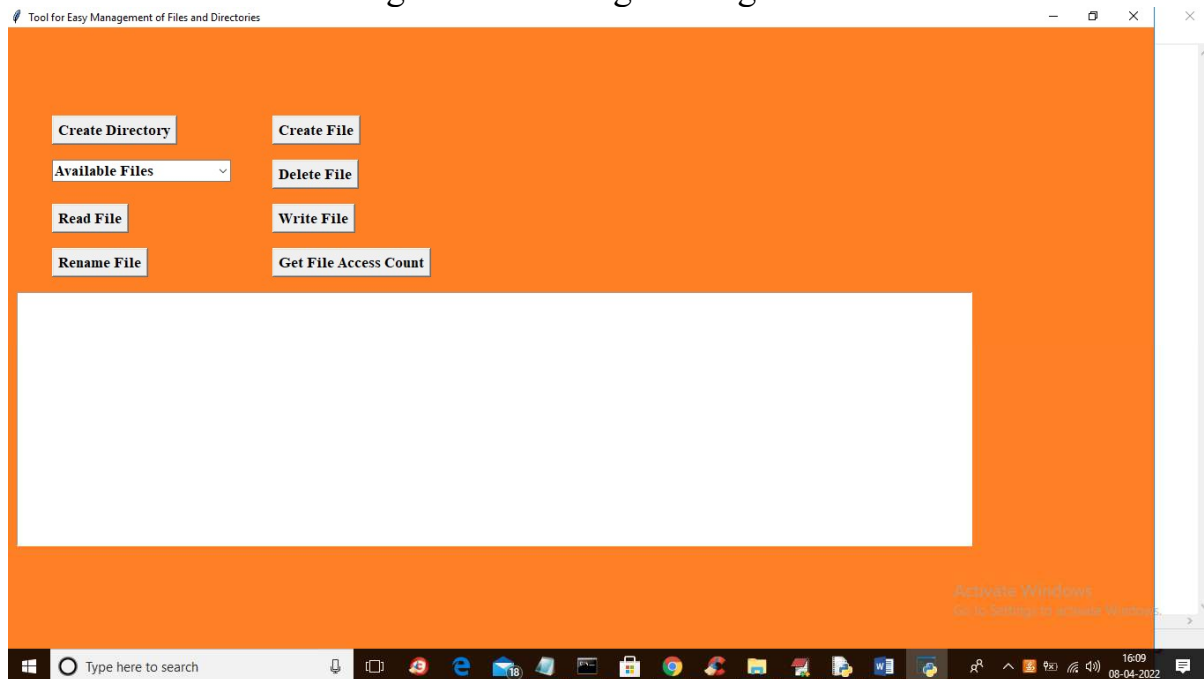


In above screen user has to enter signup details and press button to complete signup process and get below login screen



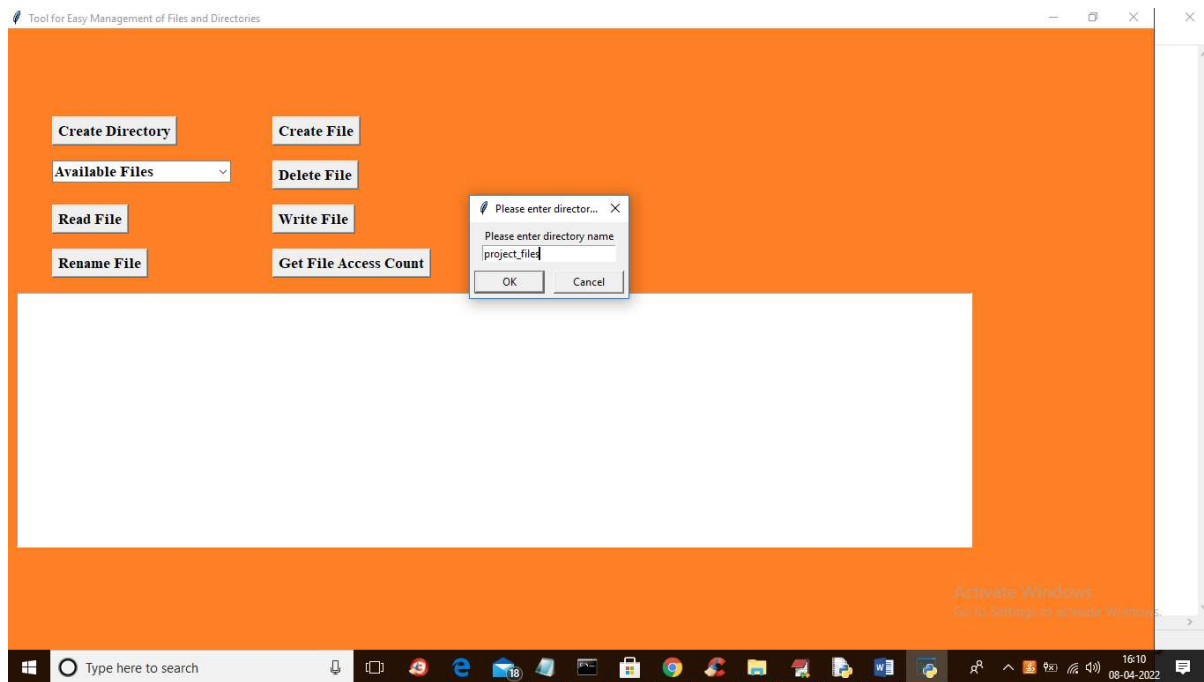
A screenshot of a web application window titled "User Login Screen". The window has a light gray background. At the top center, the text "Login Screen" is displayed in a bold, black font. Below this, there are two input fields. The first is labeled "Username" and contains the text "kumar". The second is labeled "Password" and contains six asterisks "*****". Below the password field is a button labeled "Submit".

In above screen user is login and after login will get below screen

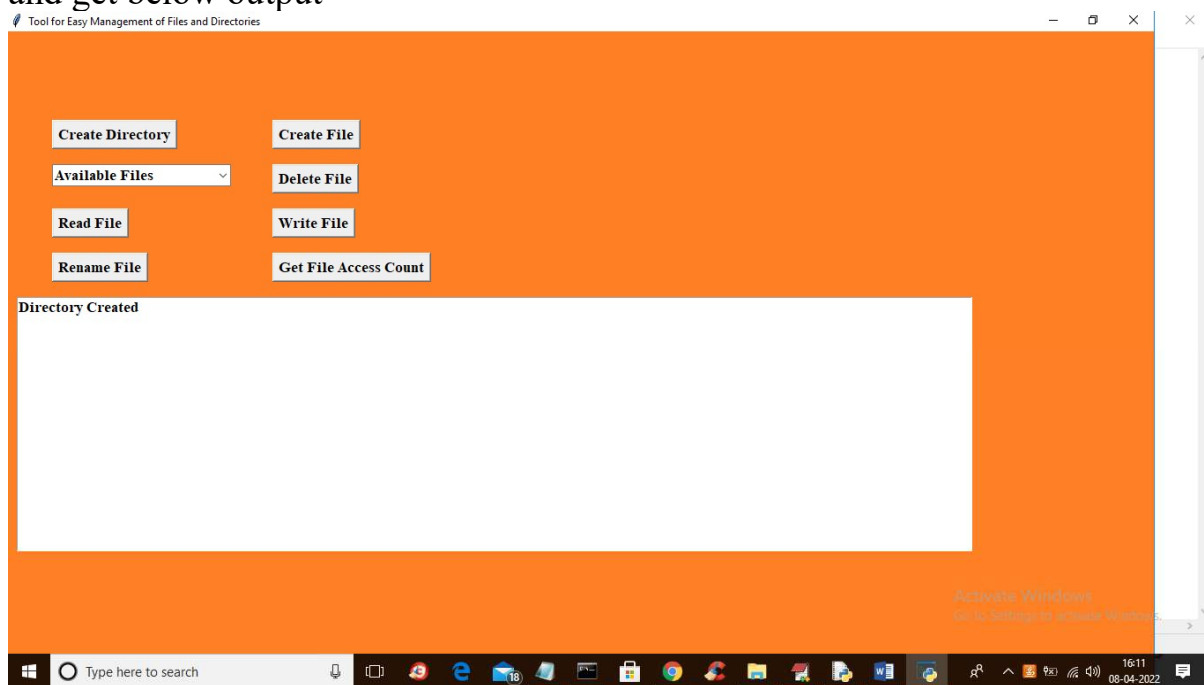


A screenshot of a web application window titled "Tool for Easy Management of Files and Directories". The window has an orange background. At the top, there are two buttons: "Create Directory" and "Create File". Below these, there is a dropdown menu labeled "Available Files" and a button labeled "Delete File". Further down, there are buttons for "Read File", "Write File", "Rename File", and "Get File Access Count". A large white rectangular area is positioned below these buttons. At the bottom of the window, there is a Windows taskbar with various icons and a search bar. The system clock shows the time as 16:09 on 08-04-2022.

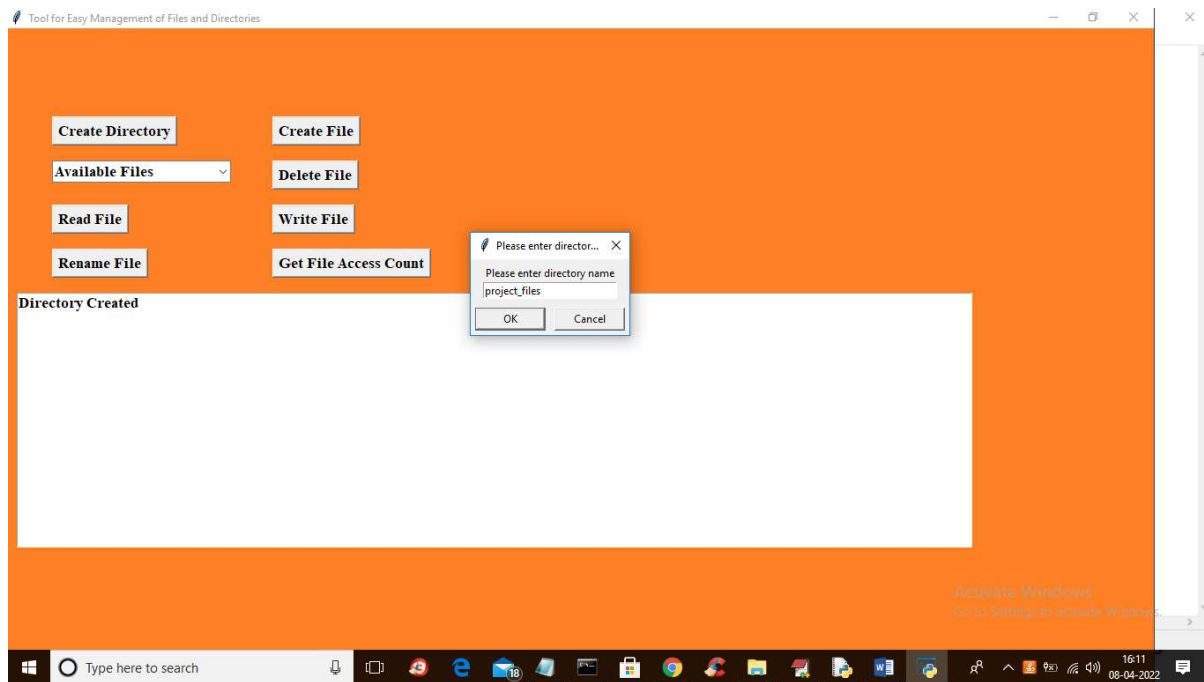
In above screen user can click on 'Create Directory' button to enter directory name and get below output



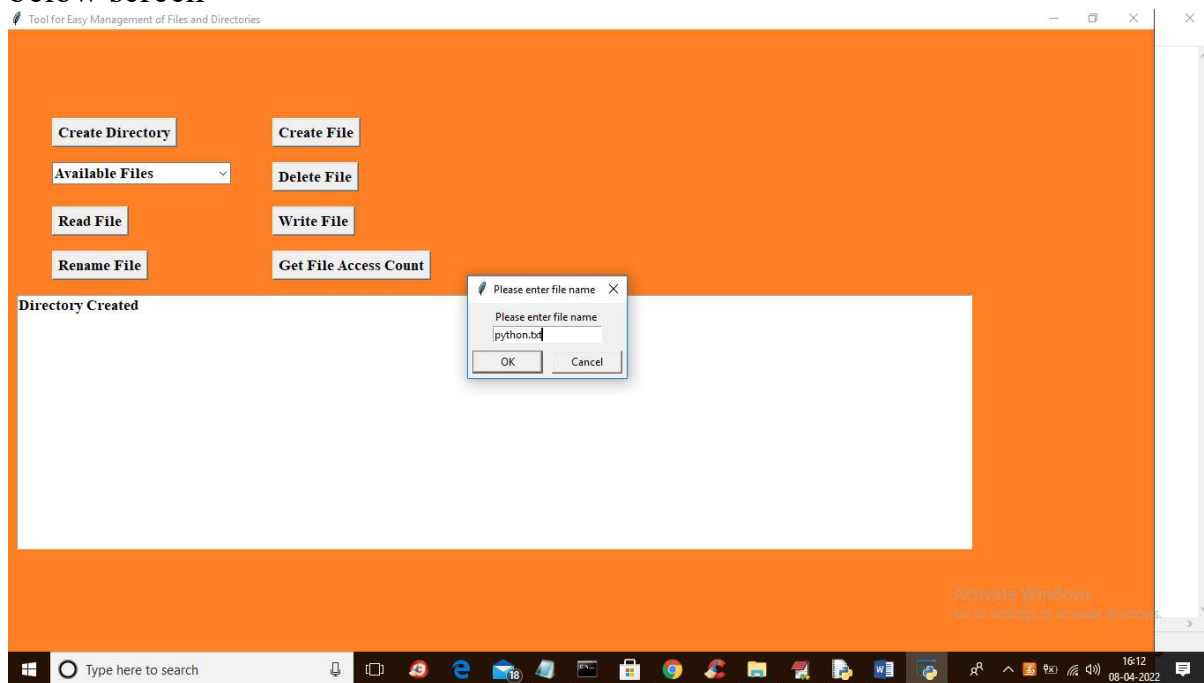
In above screen I entered directory name as 'project_files' and press OK to create directory and get below output



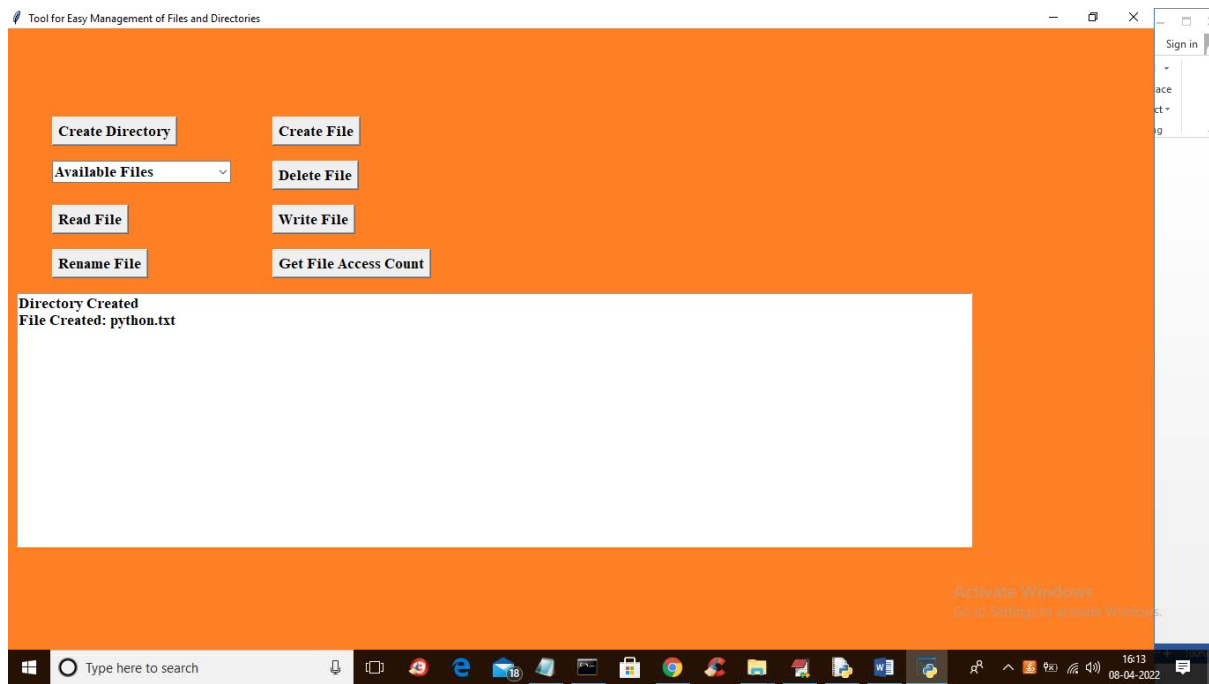
In above screen in textarea we can see directory created and now click on 'Create File' button to create file



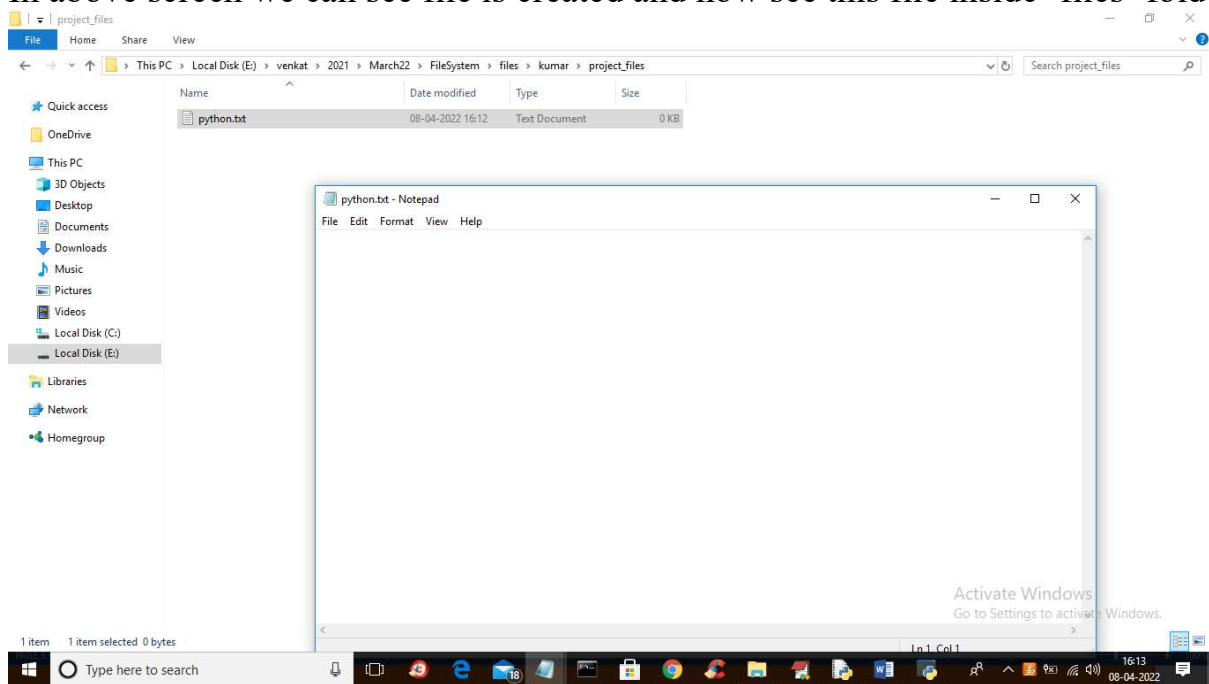
In above screen enter directory name where file has to create and then enter file name like below screen



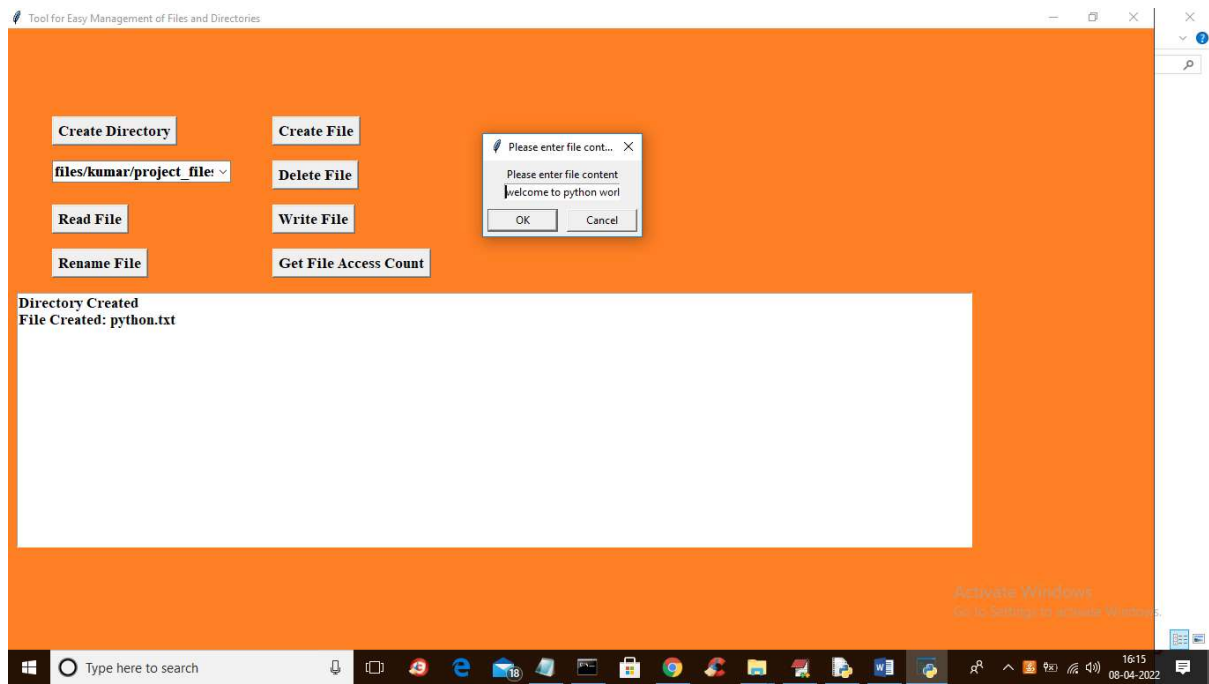
In above screen I entered file name as 'python.txt' and press OK to get below output



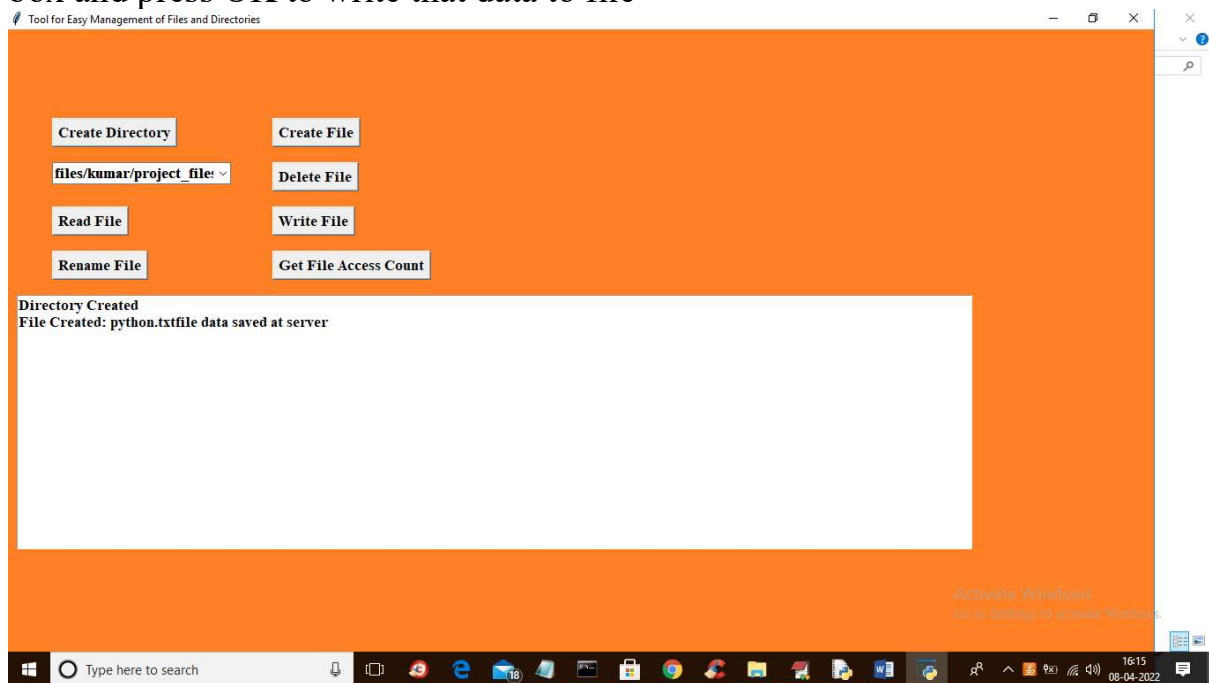
In above screen we can see file is created and now see this file inside 'files' folder



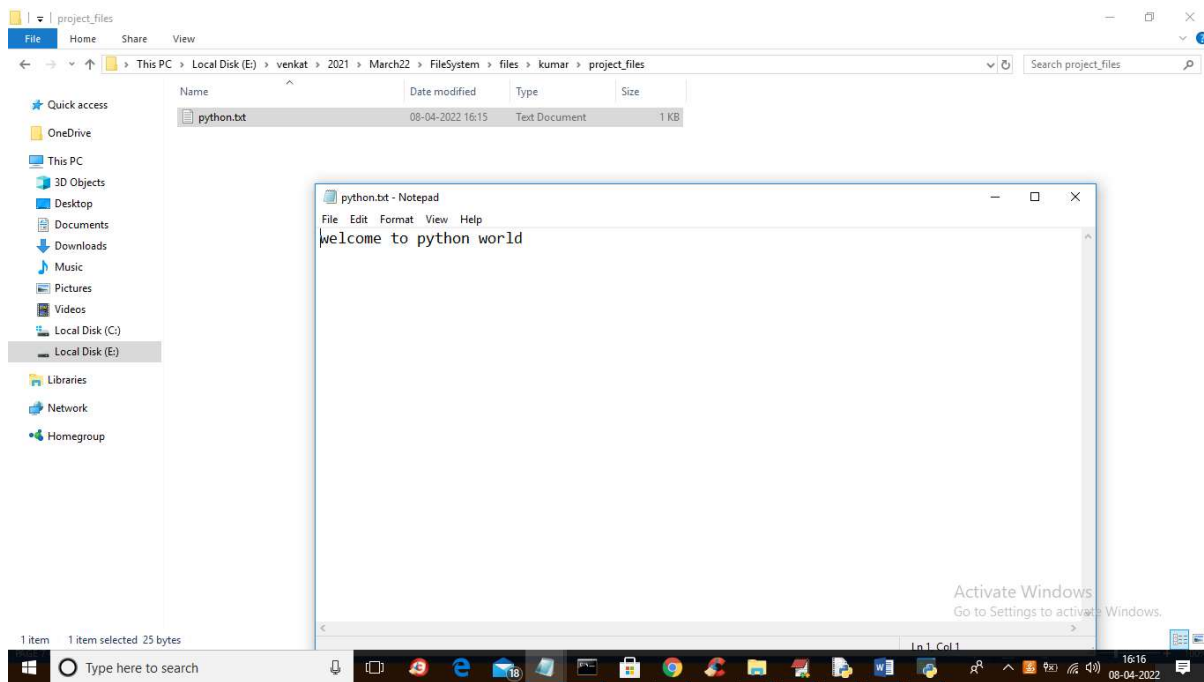
In above screen inside 'project_files' we can see python.txt file is created and now file is empty and write some text to it by clicking on write file and select that file from drop down box



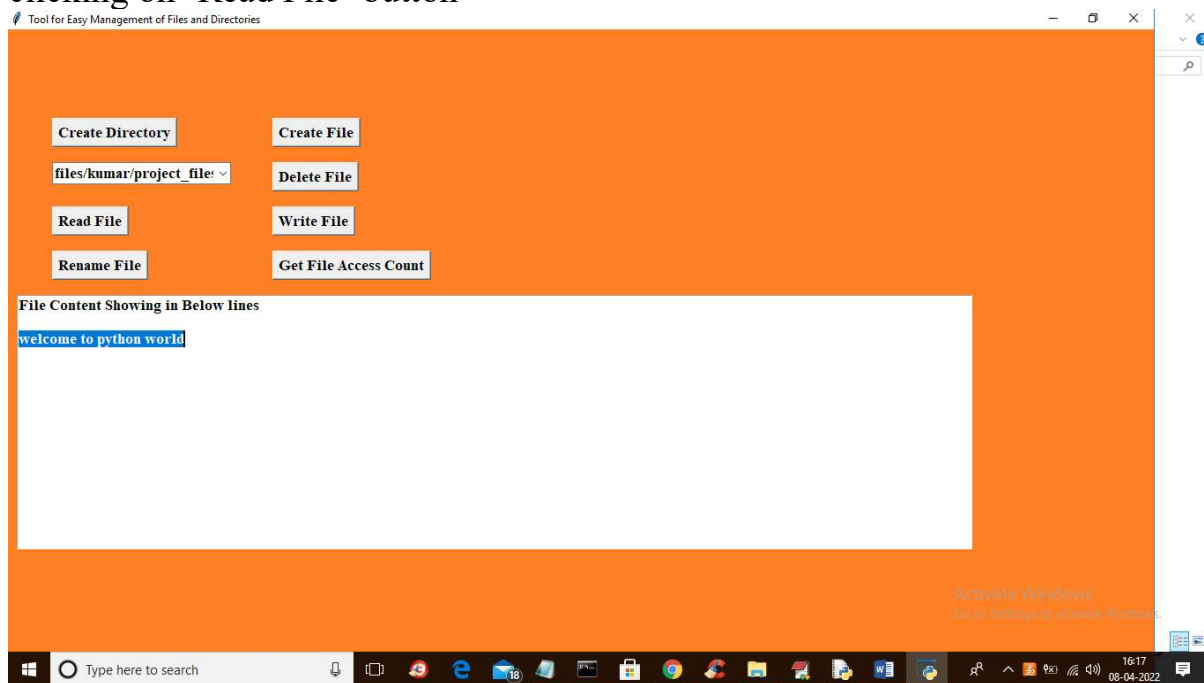
In above screen from drop down I selected file and then enter some text content in dialog box and press OK to write that data to file



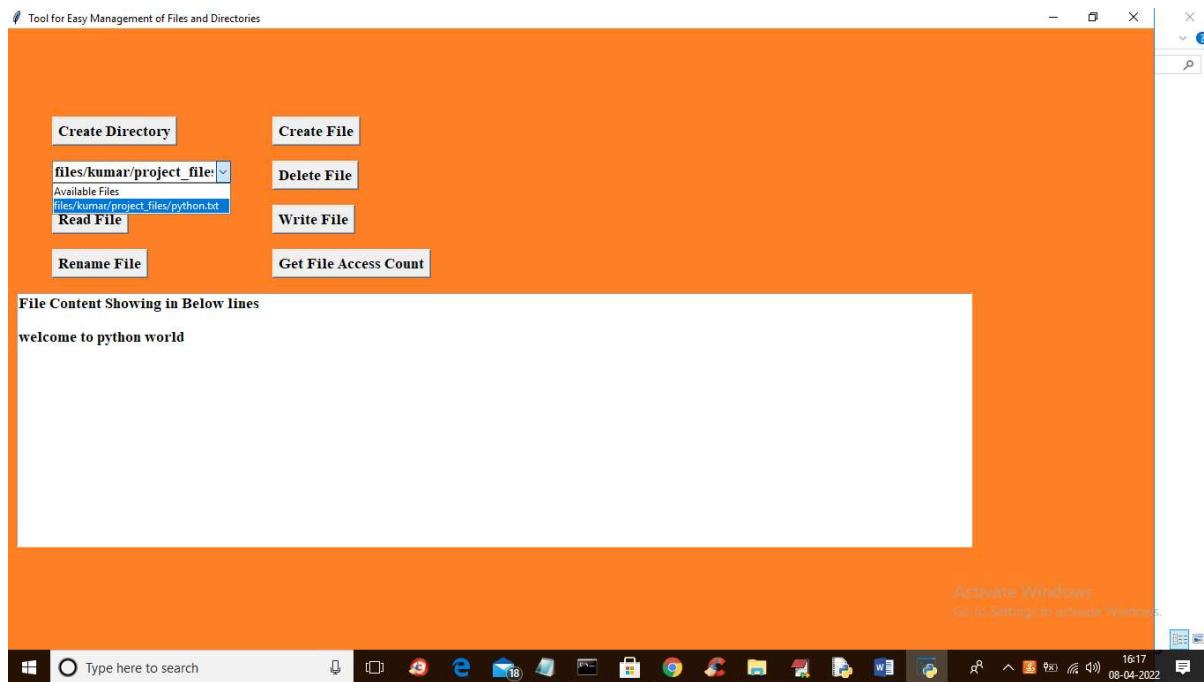
In above screen we can see data saved at file server and in files directory also we can see that text



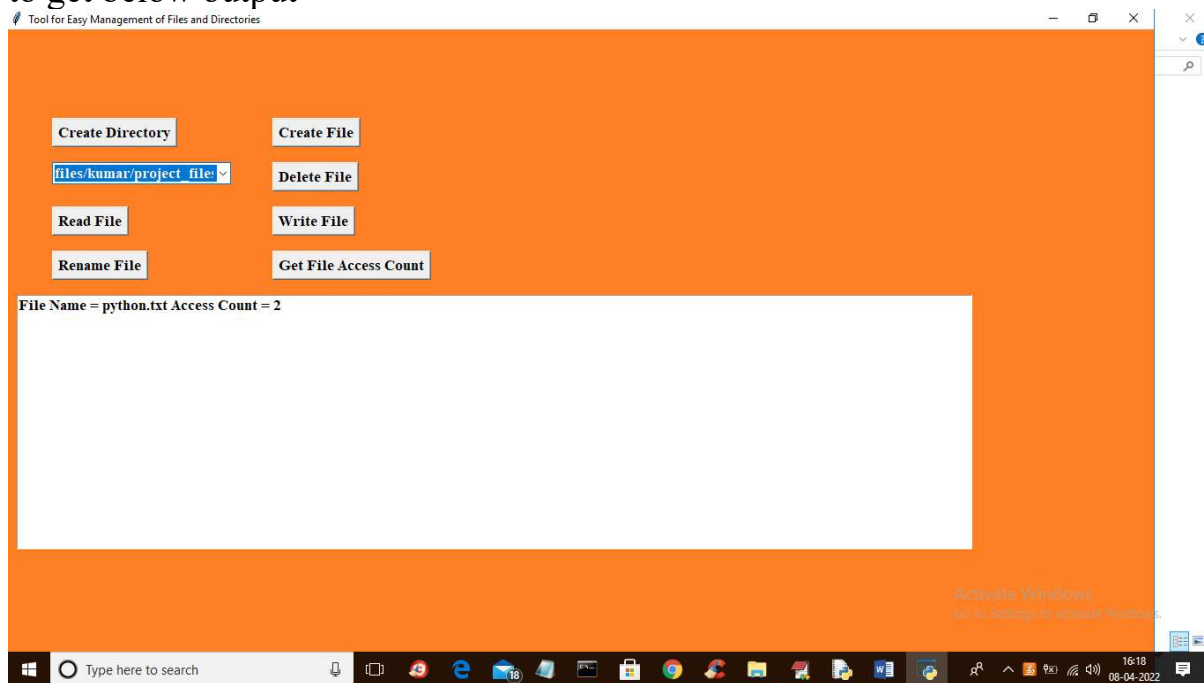
In above screen we can see some text is written to file and we can read that data back by clicking on 'Read File' button



In above screen we can see file is read and displaying to text area in blue colour and click on 'Get File Access Count' button to get access count of selected file



In above screen from drop down box I selected one file and click on 'Get File Access Count' to get below output



In above screen we can see python.txt file access 2 times and similarly you can rename and delete file

CONCLUSION

User authentication is a fundamental component in most computer security contexts. In this extended abstract, we proposed a simple graphical password authentication system. The system combines graphical and text-based passwords trying to achieve the best of both worlds. It also provides multi-factor authentication in a friendly intuitive system. We described the system operation with some examples, and highlighted important aspects of the system.

FUTURE ENHANCEMENT

When considering future enhancements for your file and directory management project in Python, you may want to broaden its functionality, improve user experience, enhance performance, and address potential limitations.

1. User Interface Improvement:

- Develop a graphical user interface (GUI) to provide a more intuitive and user-friendly experience for users interacting with your file and directory management system.

2. Search Functionality:

- Implement a search feature that allows users to search for files and directories based on keywords, file types, sizes, or creation/modification dates.

3. File Encryption and Decryption:

- Add functionality to encrypt and decrypt files to enhance security and privacy.

4. Version Control System Integration:

- Integrate with a version control system (e.g., Git) to allow users to manage versions of their files and collaborate on projects more effectively.

5. Cloud Storage Integration:

- Integrate with popular cloud storage platforms (e.g., Google Drive, Dropbox) to allow users to manage their cloud files within the application.

6. Batch Processing and Scripting:

- Provide the ability to execute batch operations or scripts on multiple files or directories, allowing for automation and streamlining repetitive tasks.

7. File Synchronization:

- Develop functionality to synchronize files and directories between different locations, devices, or cloud storage services.

8. File Metadata Management:

- Enable users to add, edit, and view metadata associated with files and directories, facilitating better organization and categorization.

9. File Compression and Decompression:

- Add support for compressing and decompressing files and directories to save storage space and facilitate easier file transfer.

10. Cross-platform Compatibility:

- Ensure that the application can run on multiple operating systems (Windows, macOS, Linux) by optimizing and adjusting the codebase as needed.

11. File Sharing and Collaboration:

- Implement features that allow users to share files and directories with others, control access permissions, and collaborate on shared files.

12. Performance Optimization:

- Optimize the application's performance for handling large files and directories more efficiently, ensuring a smooth user experience.

13. File Preview and Visualization:

- Provide the ability to preview files directly within the application, especially for common file types like images, documents, and videos.

14. Automated Backup:

- Integrate an automated backup system that allows users to schedule and manage backups of important files and directories.

15. Error Handling and Logging:

- Enhance the error handling mechanism and implement logging to track and log errors, improving the application's stability and reliability.

REFERENCE

1. Python Official Documentation:

- Python's official documentation provides comprehensive information about working with files and directories in Python.

Link: <https://docs.python.org/3/tutorial/inputoutput.html#reading-and-writing-files>

2. Real Python Tutorial:

- Real Python is a great resource for Python tutorials, and they have a tutorial on working with files in Python that you might find useful.

- Link: <https://realpython.com/read-write-files-python/>

3. GeeksforGeeks File Handling in Python:

- GeeksforGeeks offers a tutorial on file handling in Python, covering various aspects of file operations.

- Link: <https://www.geeksforgeeks.org/file-handling-python/>

4. YouTube Video - Corey Schafer's Python Tutorial - File Handling:

- Corey Schafer's tutorials are highly regarded. Check out his tutorial on file handling in Python on YouTube.

- Link: <https://www.youtube.com/watch?v=Uh2ebFW8OYM>

5. Book - "Automate the Boring Stuff with Python" by Al Sweigart:

- This book covers practical tasks in Python, including file and directory manipulation, and is great for beginners.

- Link: <https://automatetheboringstuff.com/>

6. GitHub - Python Filesystem:

- Python Filesystem is a Python module that provides an easy-to-use and intuitive way of interacting with the file system.

- Link: <https://github.com/PyFilesystem/pyfilesystem>

7. GitHub - pathlib Module:

- The `pathlib` module is part of the Python standard library and provides a more modern approach to working with file paths.

- Link: <https://docs.python.org/3/library/pathlib.html>